Arne Persyn

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EDUCATION

Howest University of Applied Sciences — Applied Computer Science Expected June 2027
Software Engineering major, currently in third year. 2025–26 average: 16.0/20 (80%), no course grade below 14/20.
Coursework: Software Architecture (19/20), System Architecture & Deployment, Secure OO Architectures

Ghent University (UGent) — Computer Science (transferred to Howest) 2022 – 2024
Coursework: Algorithms, Databases, Object-Oriented Programming

EXPERIENCE

Software engineer (Part-time) — Sempl (Wattr) Aug 2026 – Present
IoT smart pool controller managing heat pumps, pumps and water treatment for 500+ app users

- Own features end-to-end across the Go backend and Flutter mobile app, from design through release, for a platform controlling residential pool equipment.
- Built an energy-surplus heating feature: users set a maximum pool temperature in the app, and the backend automatically boosts the heat pump when surplus energy is available, increasing self-consumption and lowering heating costs.
- Designing and building an offline over-the-air firmware update flow for pumps: firmware is bundled in the app, the phone connects directly to the pump's Wi-Fi access point, and a guided step-by-step UI streams the update with live progress, enabling customers to update without an installer visit or internet connection.

Software engineer (Part-time) — Sempl Feb 2024 – May 2024

- Built a data pipeline for an external client with high-end movable-floor pools: a Python service subscribes to sensor readings over MQTT, stores pool floor height in InfluxDB, and visualizes it in Grafana.
- Instrumented the Go backend with Prometheus metrics and built Grafana dashboards tracking API request volume, giving the team visibility into API usage.
- Developed the compatible-devices page on the customer-facing Svelte website, showing customers which pool equipment works with the product.

Tennis Coach 2018 – Present

- Coached 140+ players of all ages and skill levels over eight years, planning and leading individual and group training sessions; hold two tennis coaching certifications.

PROJECTS

Howestprime — Cinema Ticketing Platform (solo) | *C#, .NET, RabbitMQ, Terraform, Azure, GitHub Actions*

- Designed an event-driven system of two .NET microservices (movies and ticketing) split along domain-driven bounded contexts, communicating through a RabbitMQ topic exchange with event contracts specified in AsyncAPI.
- Implemented an asynchronous booking and payment flow in which the movies service publishes catalog and booking events and consumes payment success/failure events to close out bookings, keeping the services fully decoupled.
- Provisioned the full Azure environment as code with Terraform, including Container Apps, Container Registry, Key Vault for secrets, managed identities, Log Analytics monitoring and the message broker.
- Built a GitHub Actions CI/CD pipeline that runs the test suite and deploys containers to Azure on every merge to main, authenticated through a dedicated service principal.

Splendor Online (team project) | *Java, HTML, CSS, JavaScript*

- Built a browser-based multiplayer version of the board game Splendor for up to 4 players, with a Java server managing game state and a JavaScript frontend.
- Implemented core backend game logic, enforcing rules and turn flow on the server.
- Designed AI bot opponents that score every legal move on six weighted heuristics (such as prestige points, card affordability and progress toward nobles), and tuned the weights with a (1+1) evolution strategy in which a champion and a mutated challenger compete across games and the better-ranked set survives.

LEADERSHIP & ACTIVITIES

Webmaster & Member — Rotaract 2020 – 2024

- Maintained the club website and online presence for 4 years and co-organized charity events raising funds for local and international causes.

SKILLS

Programming: Go, Java, Python, C#, C++, JavaScript, Dart, SQL, HTML/CSS
Frameworks: Flutter, Svelte, .NET
Tools & DevOps: Docker, Terraform, Azure, GitHub Actions, RabbitMQ, MQTT, InfluxDB, Prometheus, Grafana, Firebase, Git
Languages: Dutch (native), English (fluent)